

Construct Validity for Cross-Context Consistency Scores: Confound Tests Across 36,000 Drug, Genetic, and Morphological Perturbations

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Abstract

Background: Given thousands of profiled compounds and budget for dozens, a practitioner ranks by a cross-context consistency score and follows up the top of the list. Direction instability, which measures the angular consistency of a perturbation’s response across contexts—cell lines, cell types, or imaging sites—produces such rankings. Whether they survive correction for toxicity programs, off-target consistency and universal-essential effects has not been tested.

Results: We report eight pre-registered confound tests across 10,625 drug and genetic perturbations in LINCS L1000 and Perturb-seq, with Holm–Bonferroni correction within domain, and a morphological feature-ablation sensitivity analysis in 24,992 JUMP-CP compounds. The ranking is stable in aggregate: raw and corrected scores agree at Spearman $\rho = 0.91$ in Perturb-seq and $\rho > 0.99$ in LINCS, and the morphological ablation leaves the JUMP-CP ranking nearly unchanged ($\rho = 0.998$, $n = 24,992$). It is not stable item by item: proteasome and spliceosome components move 660–900 positions of 1,676 under a correction whose global effect is $\rho = 0.91$. The displacement is not noise. Target-axis projection tracks signless alignment to the annotated target’s shRNA axis ($\rho = 0.317$, versus -0.058 for the raw score), a pattern an independent CRISPRi reference reverses, and separates compounds the raw score cannot: daunorubicin and MG-132 have near-identical scores (0.557, 0.565), but only daunorubicin’s consistency projects onto its annotated target. Universal-essential knockdowns diverge across cell types rather than converging on a shared lethal response ($d = 0.60$, $p_{\text{perm}} < 10^{-4}$), the opposite of the naive confound expectation. Toxicity correction shifts scores by less than 0.01, refuting the predicted confound: cytotoxic compounds already occupy the high-instability regime.

Conclusions: Corrections leave the population ranking largely intact; they identify which perturbations move under correction. A score adequate for population-level comparison is not adequate for a claim about any single perturbation, and a high global correlation between raw and corrected rankings is not evidence that the correction was unnecessary.

Keywords: construct validity, geometric perturbation scores, mechanistic claims, direction instability, LINCS L1000, Perturb-seq, Cell Painting, rank stability, pre-registration, Fréchet variance, Grassmannian geodesic

1 Background

Geometric perturbation scores—metrics that quantify how a biological system’s response to a perturbation varies across experimental contexts—rank drugs, genes, and compounds by mechanism

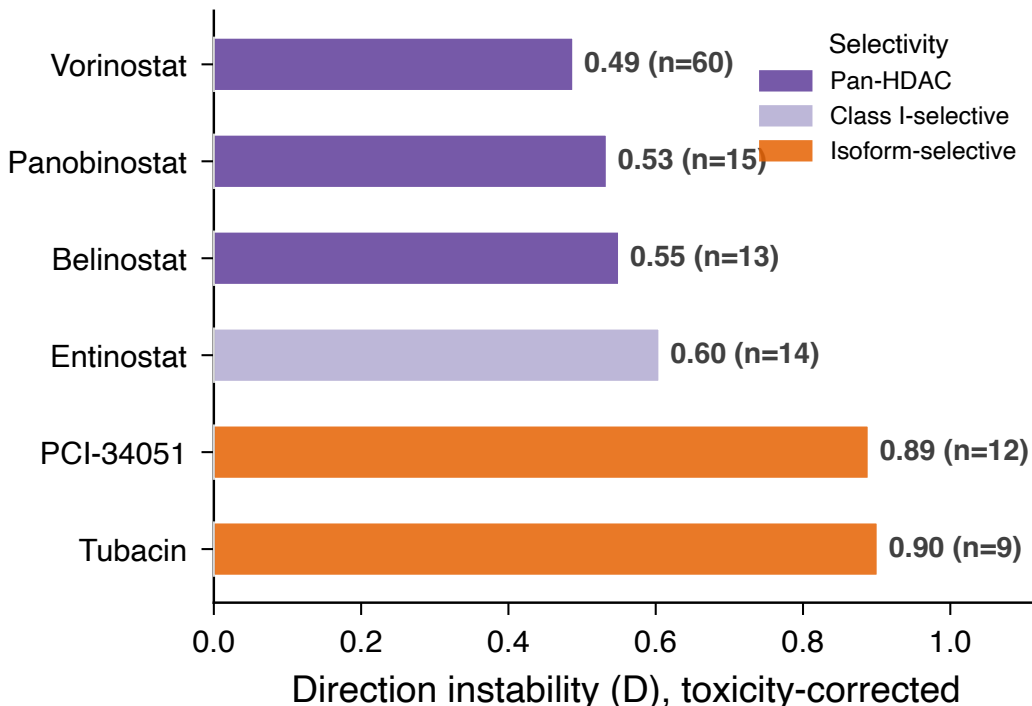


Figure 1: **HDAC selectivity gradient: calibration panel for direction instability.** Toxicity-corrected direction instability for six HDAC inhibitors, grouped by selectivity class. Pan-HDAC inhibitors [5] (vorinostat, panobinostat, belinostat) score $D = 0.49$ – 0.55 ; Class I-selective entinostat is intermediate ($D = 0.60$); the isoform-selective inhibitors tubacin (HDAC6) and PCI-34051 (HDAC8) [6] are highest ($D = 0.89$ – 0.90). Selectivity is treated as three ordered categories rather than six ranks: the pan-HDAC inhibitors differ in potency rather than in isoform breadth, and the relative selectivity of tubacin and PCI-34051 is assay-dependent, so neither pair supports an internal ordering. All 11 cross-category comparisons are concordant (Kendall $\tau_b = 0.856$). Six compounds admit no meaningful significance test, and none is reported; the panel is a calibration reference, not an external validation. Plotted values are read from `results/fig5_hdac_source/fig5_hdac_source.json`.

conservation across large-scale profiling datasets [1, 2]. Direction instability, one such score, measures the angular consistency of perturbation signatures across cell lines: a drug whose transcriptional response points in the same direction regardless of cell type receives a low score, indicating conserved mechanism [preprint, under review; 3]. Applied to 8,949 drugs in LINCS L1000, direction instability produces biologically meaningful rankings: pan-HDAC inhibitors achieve the lowest scores ($D = 0.49$ – 0.55), with a selectivity gradient within the HDAC class (Figure 1) that tracks isoform specificity [preprint, under review; 4]. Those two studies establish the score and its behavior at scale. Whether its rankings survive correction for the confounds that motivate them is the question here.

Direction instability complements rather than replaces the perturbation-quality and retrieval metrics already in use on these datasets, and the estimands differ. The L1000 Transcriptional Activity Score combines signature strength with replicate concordance to identify transcriptionally active perturbations, so it registers the conjunction that D separates. Cell Painting percent-replicating asks whether replicate profiles are more similar to each other than a null distribution allows, and

mean average precision scores retrieval of known replicate or annotation groups. Each answers whether a perturbation produced a reproducible measurable signal, or how well profiles retrieve a stated label. D answers a different question: given that a signal is present, how much its direction changes across the contexts the analyst designates. A perturbation can score well on activity and retrieval while carrying high D , and the reverse holds for a weak but directionally stable perturbation.

These rankings are consistent with the score carrying biological signal, within the limits of the calibration panel of Figure 1. They leave open a practical question: when can a practitioner trust a geometric perturbation score for downstream decisions such as hit prioritization, mechanism-of-action annotation, or cross-context prediction? A score of $D = 0.55$ means signatures point in similar directions across cell lines. It does not mean the drug acts through a specific pathway or that the observation will replicate in a new context.

Three specific confounds threaten interpretation. The raw score cannot distinguish on-target mechanism conservation from coherent off-target programs. A drug scored from few cell lines may appear consistent due to low Fréchet variance rather than genuine directional agreement. In genetic perturbation screens, universal-essential knockdowns could inflate or deflate cross-cell-type distances through a shared lethal-response subspace. No systematic procedure exists for diagnosing which of these threats a given score has survived (cf. construct-validity frameworks in measurement theory [7, 8]).

We formalize these threats through a validation framework (Figure 3): a sequence of seven conditions that progressively strengthen what a geometric perturbation score supports. Each condition targets a specific confound. The ordering reflects logical dependency: a reliable measurement (Condition 2) is prerequisite for meaningful projection (Condition 5), and localized signal (Condition 4) precedes phenotype alignment (Condition 5). Satisfying additional conditions does not change the score; it narrows the set of confounds that remain viable alternative explanations.

This paper tests whether these validation conditions matter empirically across three perturbation domains: 8,949 LINCS L1000 drug compounds [2], 1,676 Perturb-seq genetic knockdowns [9], and 24,992 JUMP-CP Cell Painting morphological profiles [10]. Direction instability serves as the case study, but the procedure—define confounds, specify corrections, pre-register predictions, test whether corrections reorganize rankings or preserve them—applies to any geometric score measuring perturbation-response consistency across contexts.

2 Definitions and Notation

2.1 Direction instability

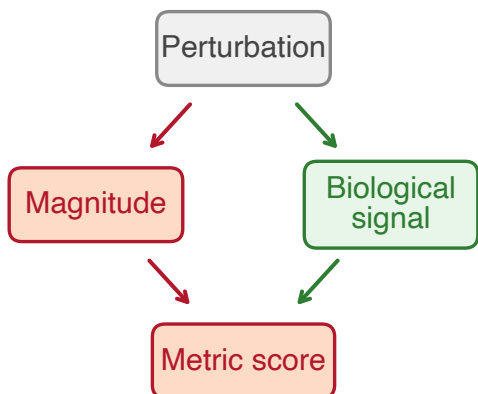
For a drug tested in K cell lines, let $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \quad (1)$$

$D = 0$ when all signatures are collinear and $D = 1$ when they are orthogonal on average, with $0 \leq D \leq K/(K-1)$, so the attainable maximum is 2.000 at $K = 2$ and 1.250 at $K = 5$.

This quantity is not new. Writing $\bar{R} = \|\sum_i \hat{\mathbf{s}}_i\|/K$ for the mean resultant length of the unit-normalized signatures, the mean pairwise cosine in Equation 1 is $(K\bar{R}^2 - 1)/(K - 1)$, which is pairwise phase consistency (PPC), introduced by Vinck et al. [11] as a bias-free measure of synchronization—so $D = 1 - \text{PPC}$. PPC was designed for the property claimed here, separating

a Confounded metrics



b Direction instability

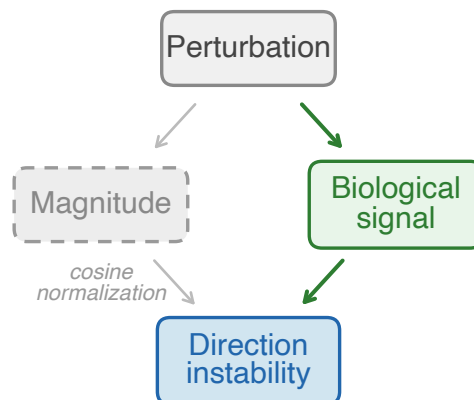


Figure 2: **Direction instability removes linear magnitude dependence.** (a) Confounded metrics (variance, subspace size) inherit effect magnitude through a shared path, producing spurious associations between perturbation strength and metric value. (b) Direction instability operates after cosine normalization, which blocks the linear magnitude path under the assumption that signature direction is independent of dose (separability). The observed association between group variable and metric reflects biological signal to the extent that this assumption holds. An empirical test of separability (dose-direction stability across 1,445 drugs) is reported in the Discussion.

consistency from apparent strength due to sample size. What this study contributes is therefore not the statistic but a construct-validity audit of it: which confounds a cross-context consistency score survives, tested against pre-registered predictions across three profiling modalities. Cosine normalization removes the linear dependence on perturbation magnitude (Figure 2), under the assumption that signature direction and magnitude are separable—that is, that the direction a drug perturbs gene expression does not itself change with dose. Direction instability measures only direction [3]. Variance-based and estimation-based metrics conflate perturbation strength with biological signal. The separability assumption is tested empirically in the Discussion.

2.2 Grassmannian geodesic distance in Perturb-seq

Direction instability generalizes from vectors to subspaces. For a gene knocked down in two cell types, let $U_1, U_2 \in \text{Gr}(k, d)$ denote the top- k PCA subspaces of the perturbation response in each cell type (k = subspace dimension, d = ambient dimension). The Grassmannian geodesic distance $\delta(U_1, U_2) = \|\theta\|_2$, where θ are the principal angles between the subspaces, measures how much the perturbation response *rotates* across cell types. With $K = 2$ contexts, the pairwise average in Eq. 1 reduces to a single comparison, and the estimand carries no Fréchet variance or holonomy information. We therefore do not compute direction instability as defined in Eq. 1 for Perturb-seq; we compute the Grassmannian geodesic distance $\delta(U_1, U_2)$ between per-cell-type response subspaces. This is a related but distinct geometric estimand, and we label it as such throughout rather than presenting it as direction instability at $K = 2$. Perturb-seq results are interpreted as a subspace-rotation analysis, not a direct replication of the LINCS vector-cosine analysis.

2.3 Direction instability in Cell Painting morphological profiles

Cell Painting [10, 12] measures 3,180 morphological features per well across five fluorescent channels. JUMP-CP profiles 116,000 compounds across 10 imaging sources. Direction instability applies directly: each compound’s morphological signature in a given source is a vector in $\mathbb{R}^{3180}$, and D measures angular consistency across sources. A JUMP-CP source is a partner imaging site rather than a distinct cell type, so D computed over sources indexes cross-site technical reproducibility rather than context-dependence across cell types, as Tower [4] reports for JUMP-CP plate replicates within a single cell line. The JUMP-CP result is reported on that basis throughout. A frozen block of 252 features (7.9% of the space) covers size and shape across compartments together with DNA- and mitochondria-channel intensity, granularity and radial distribution. The ablation removes that block and recomputes direction instability. The block is an author-defined morphological feature family rather than a validated cell-health signature; published cell-health work defines phenotypic readouts and models predicting them rather than a corresponding set of CellProfiler columns.

2.4 The validity problem

Consider two drugs with identical direction instability, $D \approx 0.56$ (mechanism-conserving regime):

- **Drug A** is a topoisomerase II inhibitor. Its conserved direction aligns with its target gene’s knockdown signature—phenotype-projected instability is high. The consistency comes from on-target DNA-damage signaling.
- **Drug B** is a proteasome inhibitor. Its conserved direction does *not* align with its annotated target subunit’s knockdown signature—phenotype-projected instability is near zero. The consistency comes from broad proteostasis disruption shared across cell types.

Both achieve $D \approx 0.56$. Both pass the raw metric equally. Phenotype projection (Condition 5 of the validation framework) distinguishes them: Drug A’s consistency is mechanistically on-target; Drug B’s is not. The distinction matters for downstream decisions: Drug A is a strong candidate for mechanism-based follow-up; Drug B requires additional investigation to identify what program drives its consistency before any mechanistic claim is supported. A concrete instance of this scenario (daunorubicin vs. MG-132) appears in the Discussion.

2.5 The validation framework

The validation framework defines seven conditions for geometric perturbation scores (Figure 3), ordered by logical dependency rather than by evidence strength:

1. **Process view declared.** What constitutes a “context”?
2. **Reliable vector field.** Adequate signal-to-noise for direction to be meaningful.
3. **Perturbation estimable.** Effect isolated from background.
4. **Localized to biology.** Instability concentrates in pathway-relevant genes.
5. **Phenotype-aligned.** Conservation direction matches a known target pathway.
6. **Transports across contexts.** Score replicates in held-out contexts (low Fréchet variance).
7. **Predicts holdout.** Metric predicts an external outcome in new data.

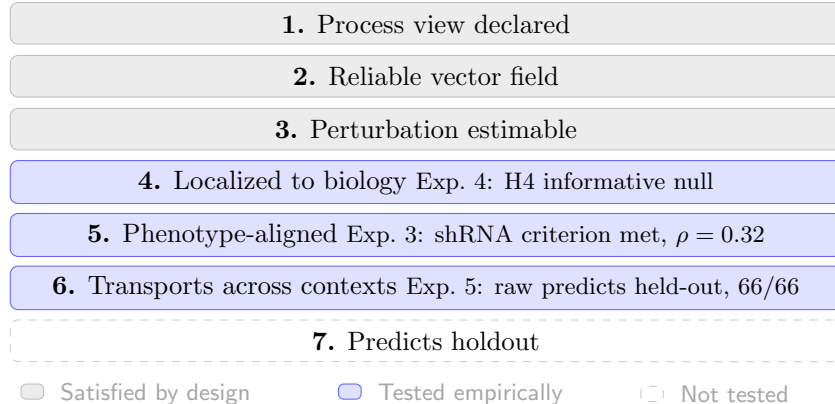


Figure 3: **The validation framework.** Seven conditions progress from measurement prerequisites (1–3) through empirically tested validation conditions (4–6) to external prediction (7). Each condition targets a specific confound. Conditions 4–6 are tested across three domains (LINCS L1000, Perturb-seq, JUMP-CP); result summaries are shown at right. Satisfying additional conditions does not change the direction-instability score—it narrows the set of confounds that remain viable.

Conditions 1–3 are satisfied by design in our datasets and are not tested further. Conditions 4–6 are tested empirically. Condition 7 (holdout prediction in a genuinely new dataset) is not tested: Experiment 5’s leave-one-cell-line-out cross-validation tests Condition 6 (transport within LINCS), not generalization to an independent dataset. A Condition 7 test would require predicting compound activity in a dataset collected independently of LINCS—such as a proprietary pharma screen or a future LINCS phase—using only validation-framework scores computed from the current data.

3 Related work

Validation frameworks for computational scores. Recent work has formalized validation procedures for computational scores: construct-validity frameworks for ML benchmarks [13, 14], the SBB principles for perturbation-prediction models [15], and the classical construct-validity tradition in psychometrics [7, 8]. Our validation framework draws on these precedents, instantiating them for geometric perturbation scores. A mapping between the framework’s conditions and Messick’s six validity facets appears in Supplementary Table S1.

Perturbation biology. CERES [16] corrects copy-number artifacts in CRISPR screens; integrated Broad–Sanger analyses [17] document cross-study divergence. These corrections operate on dependency-score magnitudes; our contribution is orthogonal, correcting subspace geometry and showing that rank-preserving corrections carry diagnostic value. The Connectivity Map framework [1, 2] and subsequent metric evaluations [18] establish the transcriptomic-similarity scoring approach that direction instability extends. LEMUR [19] performs multivariate regression on Grassmann manifolds for multi-condition Perturb-seq; our Experiment 7 uses a simpler Grassmannian geodesic distance. Raju’s geometric coherence framework [20] applies related angular measures to single-cell CRISPR perturbations; our work differs in testing construct validity of such scores rather than using them directly for prediction. Pharmacogenomic interaction landscapes [21] and retrieval-based profile evaluation [22] provide complementary perspectives on cross-study consis-

tency.

Metric evaluation in perturbation biology. Prior work has benchmarked connectivity and signature-similarity metrics for consistency and predictive value [2, 18, 23], validation frameworks have been proposed for ML benchmarks and computational scores [13, 14]; and recent work has shown that simple linear baselines match deep-learning perturbation predictors [24], while neural optimal transport [25] offers an alternative geometric approach to perturbation response modeling. These efforts evaluate *whether* a metric predicts a downstream label. Our contribution is orthogonal: we test *which confounds a given score has survived*—treating validation as a graded, condition-dependent diagnostic rather than a single predictive-accuracy number—and identify where those conditions affect interpretation across three perturbation modalities. We are not aware of a comparable condition-by-condition validation applied to an unsupervised geometric perturbation score with pre-registered decision criteria.

4 Methods

4.1 Data

Perturbation signatures were drawn from LINCS L1000 Phase I (GEO accession GSE92742; ~ 1.3 million profiles across ~ 50 cell lines), using Level 5 replicate-collapsed z-scores across 978 landmark genes [2, 26]. Of 77 cell lines in Phase I, approximately 50 contribute sufficient compound coverage for direction instability computation. We retained 8,949 small-molecule perturbagens tested in ≥ 5 cell lines.

Perturb-seq data derive from Replogle et al. [9]: genome-scale CRISPRi knockdowns in K562 and RPE1 cells. We extracted 1,676 genes with ≥ 30 perturbed cells in both cell types, computed per-gene expression responses over 2,073 highly variable genes, and fitted $k = 5$ PCA subspaces per gene per cell type (of these, 1,567 have matching DepMap Chronos labels for HP3; all 1,676 are used for HP1). Both screens target DepMap-essential genes; 85% of RPE1 targets overlap with K562 targets. Essentiality labels derive from DepMap Chronos scores [27]: 855 genes with Chronos < -0.5 in both K562 and RPE1 were classified as universal-essential; the remaining 712 are sub-threshold and serve as the comparison group. A label-permutation null (10,000 permutations) tested whether the observed group difference exceeds background K562/RPE1 divergence.

JUMP-CP Cell Painting profiles derive from the Cell Painting Gallery [10]: 24,992 compounds tested in ≥ 5 imaging sources after filtering (78.4% attrition from the 115,729 total; the majority of compounds appear in fewer than 5 sources, making direction instability—which requires multiple contexts—undefined for them), with eight plate-level control compounds excluded. The ablated block (252 features matching Area, Compactness, Eccentricity, Extent, FormFactor, Solidity, DNA/Mito Intensity, Granularity and Radial Distribution) was resolved once against the release schema and hashed; ablated direction instability was computed on the remaining 2,928 features.

Following pre-registration principles [28], all scoring functions and drug lists were committed before any real LINCS signatures were analyzed (initial freeze: commit 8d74bd0). Perturb-seq hypotheses and extended drug hypotheses (H5-full, H3-CRISPRi) were pre-registered before the correction analysis (commit f3c88ed, tag prereg-extended-psb, timestamped 2026-07-08T11:56:58–04:00, preceding experiment results committed at 14:42–04:00). The dose-direction separability test (Section 6.2) was pre-registered in commit 5bb6080 before execution. The repository was public

Table 1: Deviation log. All departures from the pre-registered plan, with rationale and timing. Full detail in `DEVIATION_LOG.md`, commit-linked.

#	Pre-registered	Actual	Why	When
1	Hypotheses frozen; gene-set method unspecified	Gene-set method declared in extended pre-reg	Original freeze omitted gene-set construction parameters for H3/H4/H5	Pre-analysis
2	H5: $\geq 4/5$ folds win	Threshold adjusts to $[0.8n]$ if < 5 valid folds	Maintains 80% win rate when rare cell lines have too few drugs; not triggered in practice (71 cell lines)	Pre-analysis
3	H3 and H4 treated as independent	Shared shRNA-consensus dependency documented	Both derive from the same genetic perturbation object; not independent tests	Pre-analysis
4	Original H5: 5 alphabetical folds	Full 66-fold leave-one-out with Wilcoxon co-primary	Extended pre-reg (commit <code>f3c88ed</code>) replaces original	Pre-analysis
5	H3-CRISPRi: three-outcome decision tree	Projected $\rho = 0.54$; raw $\rho = -0.19$	Decision tree followed; the registered K562-only confound explanation is not applied (<code>DEVIATION_LOG.md</code> entry 12)	Post-analysis
6	H5 win test $\rho_{TS} > \rho_{raw}$	Reported as not confirmed; predictive strength compared on $ \rho $	The registered test compares signed correlations between oppositely-oriented statistics, so its criterion is satisfiable by orientation rather than by prediction; the registration omitted an orientation convention	Post-analysis

at the time of all freezes.¹

Reporting guidelines. No existing reporting guideline (such as TRIPOD for clinical prediction models) directly governs a validation study of an unsupervised geometric score. We adopt TRIPOD’s spirit where applicable—pre-specified analysis plan, frozen thresholds, complete reporting of all pre-registered hypotheses including nulls—and note deviations in Table 1. We omit items (such as patient-level calibration and outcome incidence) that do not apply to a non-clinical perturbation-screen setting.

We treat the five LINCSeq hypotheses (H1–H5) as one test family and the two Perturb-seq hypotheses carrying pass/fail thresholds (HP1–HP2) as a second. The separation reflects genuine

¹Code, pre-registration, deviation log, and all experimental scripts: <https://github.com/elliotttower/direction-instability-drug-validity>. LINCSeq data: GEO GSE92742; Perturb-seq: Replogle et al. [9]; JUMP-CP: Chandrasekaran et al. [10]; DepMap: Dempster et al. [27].

independence: the two domains use different experimental systems (50+ cell lines vs. two specific lines), different perturbation types (small molecules vs. CRISPRi), different geometric estimands (vector cosines in $\mathbb{R}^{978}$ vs. Grassmannian geodesics in $\text{Gr}(5, 2073)$), and share no data or parameters. A single family across all eight hypotheses would be more conservative; we consider it overly so given the absence of any shared data or analytic dependence between domains. The choice is immaterial for the confirmed result: H3 ($p < 10^{-27}$) survives Holm–Bonferroni correction even under a single 8-hypothesis family. Experiment 8 enters no multiplicity family: it tests no registered hypothesis, and its context axis is the imaging source rather than a biological one. After Holm–Bonferroni correction within the drug family, H3 ($p < 10^{-27}$) survives at $\alpha = 0.05$; H2 ($p = 0.28$), H4 (gap = 0.018) and H5 (10 of 66 folds) do not; H1’s criterion-pass is an artifact. HP3 registered no pass/fail threshold, so it enters no multiplicity family and is reported descriptively.

4.2 Phenotype-projected instability (Experiment 3)

For each drug with an annotated target gene (Drug Repurposing Hub [29], 1,551 drugs), the “target direction” is the consensus shRNA knockdown signature of that target gene in LINCS L1000 (4,369 genes eligible, ≥ 3 hairpins each, consensus averaged across up to 20 cell lines). Phenotype projection projects the instability vector onto this target direction. Signless target-axis alignment is defined as $\cos^2(\bar{\mathbf{s}}_{\text{drug}}, \bar{\mathbf{s}}_{\text{shRNA}})$: the squared cosine between each drug’s mean signature and its target gene’s shRNA consensus, measuring how much of the drug’s transcriptomic effect aligns with the genetic perturbation of its annotated target. The squared cosine is signless: it cannot separate alignment from anti-alignment.

shRNA signatures have well-documented off-target effects [30], which add noise to the target direction and attenuate the measured association. The shRNA consensus averages across up to 20 cell lines, producing a cell-type-agnostic target direction at the cost of off-target contamination. Similarly, Drug Repurposing Hub target annotations carry noise from multi-target compounds and legacy assignments; misannotation further attenuates the correlation. Polypharmacology compounds this: drugs with strong secondary targets may show phenotype projection onto the primary target even when cross-cell-line consistency is driven by a secondary mechanism, a confound that single-target annotations cannot resolve. A convergent-validity check using CRISPRi signatures (single cell type, cleaner loss-of-function but cell-type-specific) is reported in the Results.

The pre-registered prediction (H3) required Spearman $|\rho| > 0.3$ between phenotype-projected instability and signless target-axis alignment, and $|\rho| < 0.15$ for raw direction instability against the same quantity.

4.3 Variance-regularized direction instability (Experiment 5)

Variance-regularized direction instability (TS) adjusts the raw score by subtracting a penalty proportional to the Fréchet variance of the signature directions across contexts:

$$\text{TS} = D - \lambda \cdot \text{Var}_{\text{Fréchet}} \quad (2)$$

where $\text{Var}_{\text{Fréchet}} = \frac{1}{K} \sum_i \arccos^2(\hat{\mathbf{s}}_i^\top \bar{\mathbf{s}})$ and $\bar{\mathbf{s}}$ is the Fréchet mean direction. The alternative (a ratio) would be undefined when $\text{Var}_{\text{Fréchet}} = 0$ and would amplify noise for highly consistent drugs. The penalty weight λ was fixed at the value from Tower [3] and was not optimized for this evaluation; a sensitivity analysis across a 20-fold range of λ values is reported in Section 6.5.

We evaluate via leave-one-cell-line-out cross-validation over all cell lines with ≥ 20 eligible drugs after excluding the held-out cell line (66 valid folds out of 71 total; 5 excluded for insufficient drug count). A preliminary 5-fold analysis using alphabetically-selected cell lines (A375, A549,

A673, AGS, ASC) established the rare/common regime distinction with permutation nulls; the full leave-one-out extends this to all cell lines and adds Wilcoxon signed-rank as a co-primary criterion.

The pre-registered prediction (H5) required TS to outpredict raw direction instability in $\geq 80\%$ of valid folds (co-primary 1), with Wilcoxon signed-rank on per-fold $\Delta\rho$ rejecting H_0 : median ≤ 0 at $\alpha = 0.05$ (co-primary 2).

4.4 Essential-gene correction on Perturb-seq (Experiment 7)

We computed the Fréchet mean subspace of the 855 essential-gene perturbation responses (separately in K562 and RPE1) via iterative projection on $\text{Gr}(5, 2073)$ (100 iterations, convergence to $\Delta < 10^{-6}$ in Grassmannian distance; the result is insensitive to initialization because the essential-gene subspaces cluster tightly). We then projected this essential-response subspace out of each gene’s response subspace, re-orthogonalized, and recomputed geodesic distances.

Two pre-registered predictions were tested. HP1 required Spearman ρ between raw and corrected distances below 0.70 (ranking-disruptive) or ≥ 0.83 (ranking-preserving): a correction reorganizes the population ranking at $\rho < 0.70$, and the ranking survives the correction at $\rho \geq 0.83$. The ranking-preserving threshold of 0.83 was calibrated to the within-class ordering of the HDAC selectivity gradient (Figure 1) and is retained as the pre-registered decision rule. It rests on six compounds and is not offered as a generally validated cutoff; every domain-level classification is unchanged at 0.89 or 0.90, since the observed correlations are 0.91 and >0.99 . The ranking-disruptive threshold of 0.70 was set as a round-number boundary below which more than half the variance is unexplained ($R^2 < 0.49$), following the companion paper [3]. Values between 0.70 and 0.83 would require further analysis to interpret. HP3 predicted that universal-essential knockdowns produce higher geodesic distances than non-essential knockdowns (Cohen’s $d > 0$). The registration records this hypothesis as descriptive, with no pass/fail threshold, and discloses that the direction had been observed on a preliminary run before the plan was frozen. HP3 therefore registers the robustness of an effect already seen, not its direction, and is reported as directional support rather than as a confirmed prediction.

4.5 Morphological feature ablation on JUMP-CP (Experiment 8)

Experiment 8 is a sensitivity analysis and carries no registered hypothesis. It asks whether the compound ranking produced by D survives deletion of a structured block of morphological coordinates. The 252 ablated features are 7.9% of the space, against 4.2% (41/978) for the LINCS toxicity gene set.

Deleting 252 of 3,180 coordinates perturbs a cosine whatever the coordinates are, so the block is compared against 200 ablations of randomly selected feature sets of the same size rather than against no deletion. Robustness is assessed by dropping each imaging source in turn and by summarizing absolute percentile-rank shifts. Eight compounds present on at least half of all eligible COMPOUND plates are plate-level controls rather than screened compounds and are excluded before any ranking is formed.

The 0.70 and 0.83 thresholds are reported for continuity with the registered domains. They do not discriminate here: random deletion of the same number of features gives a median ρ of 0.9998, so any ablation of this size clears 0.83.

4.6 Supplementary experiments

Three additional drug-domain experiments are reported in Supplementary Experiments. **Experiment 1** (H1, toxicity correction) tests whether cytotoxic drugs masquerade as mechanism-

conserving. **Experiment 2** (H2, broad-mechanism drugs) tests whether broad-target therapeutics fall below the population median. **Experiment 4** (H4, localization score) tests whether gene-set restriction improves MOA prediction. HP2 (GO pathway prediction from corrected Perturb-seq distances) is also in Supplementary Experiments.

5 Results

5.1 Pre-registered scorecard

Table 2 summarizes the eight pre-registered hypotheses, with the JUMP-CP sensitivity analysis listed separately beneath them. One of five drug hypotheses survives Holm–Bonferroni correction (H3); H2 meets its pre-registered criterion but does not survive multiplicity correction; H1 is refuted with an artifact criterion-pass; H4 is an informative null. In the Perturb-seq domain, HP3 gives directional support; HP1 shows the population ranking survives the correction; HP2 is a null. Experiment 8 carries no registered hypothesis and appears in the table as a sensitivity analysis: the morphological ablation preserves the JUMP-CP ranking, and the registered threshold does not separate that from deleting an equal number of arbitrary coordinates.

5.2 Held-out cell-line agreement (Experiment 5)

H5 is not confirmed. The registered criterion was met (66/66 folds) but is uninformative: direction instability measures inconsistency while held-out cosine measures agreement, so a working score must correlate negatively, and the variance-regularized variant subtracts a term that reverses that sign. Compared on predictive strength, the variant wins 10 of 66 folds and loses in the two largest ($|\rho| = 0.155$ against 0.480 for A375; 0.201 against 0.494 for A549). Five cell lines were excluded for having < 20 eligible drugs.

The comparison establishes a property of the raw score instead. Direction instability computed on the retained cell lines predicts whether the held-out cell line agrees with their consensus: mean Spearman $\rho = -0.602$ (range $[-0.784, -0.223]$), correctly signed in all 66 folds. A drug whose training signatures disagree with one another also disagrees with a cell line it has not been measured in. Tower [3] reports the same prediction on this dataset under a different estimator and fold structure (AUROC = 0.986 against a permutation null of 0.500 ± 0.012); the leave-one-cell-line-out result here recovers it as a rank correlation. The advantage is drug-specific rather than algebraic. Under a variance-preserving permutation null on the preliminary five-fold analysis, held-out *signatures* are shuffled across drugs while training data remain fixed, and held-out cosines are recomputed against each drug’s original consensus. Across 1,000 permutations the observed prediction falls entirely outside the null distribution ($p_{\text{perm}} < 10^{-3}$).

Prediction is strongest in cell lines with many profiled drugs and weakest in rare ones, where removing a single cell line substantially changes the consensus direction. Across the 66 folds ρ ranges from -0.784 to -0.223 , and the folds with the fewest eligible drugs sit at the weak end of that range. How many contexts a ranking requires before it reproduces is the subject of separate work.

5.3 Target-axis projection under genetic references (Experiment 3)

Among 795 drugs with annotated targets and matched shRNA consensus signatures, phenotype-projected instability correlates with signless target-axis alignment at Spearman $\rho = 0.317$ ($p =$

Table 2: Pre-registered scorecard. *Criterion* = did the result meet the frozen pre-registered threshold? *Holm* = does it survive Holm–Bonferroni correction within its domain family ($\alpha = 0.05$)? *Status* = the interpretive status after both checks. H1 and H2 dissociate: both meet their pre-registered criteria, yet H1 is refuted (criterion-pass is an artifact) and H2 does not survive multiplicity correction. Result details for starred entries appear in Section 5.6. The corrected registered criterion was met against the LINCS shRNA reference; convergent validity against the independent CRISPRi reference remains unresolved.

	Prediction	Result	Criterion	Holm	Status
H1*	Toxicity reclassifies	$\Delta < 0.01$	Met (artifact)	—	Refuted —artifact of pre-existing high instability
H2*	Broad drugs below median	7/10 below	Met	No	Suggestive —does not survive multiplicity
H3	$ \rho_{\text{proj}} > 0.3$	$\rho = 0.317$	Met	Yes	shRNA criterion met
H4*	AUROC gap > 0.05	Gap = 0.0005	Not met	No	Informative null
H5	TS wins $\geq 80\%$ folds	66/66 on signed ρ ; 10/66 on $ \rho $	Met (artifact)	—	Not confirmed
HP1	Ranking not preserved ($\rho < 0.70$)	$\rho = 0.91$	Not met	—	Ranking preserved
HP2*	GO prediction improves	$\Delta\rho = -0.02$	Not met	—	Null
HP3	Essential $d > 0$ (descriptive)	$d = 0.60$	—	—	Directional support —inverts the predicted confound
<i>Unregistered sensitivity analysis</i>					
Exp. 8	—	$\rho = 0.998$, $n = 24,992$	—	—	Ranking preserved

4.8×10^{-20}). Raw direction instability is not detectably associated with alignment ($\rho = -0.058$, $p = 0.10$). Both pre-registered criteria are satisfied.

Raw direction instability measures whether a drug’s signatures point in consistent directions, and that consistency could arise from any coherent program. Projection restricts the dispersion to the axis defined by the annotated target’s shRNA consensus. Because projected dispersion and alignment are both built from that axis, their association does not by itself establish that projection isolates a mechanistically specific component. In the shRNA arm, raw direction instability was not detectably associated with alignment, and under the K562 CRISPRi reference it is negatively associated with alignment.

The gene-set construction avoids researcher degrees of freedom: the shRNA consensus is an empirical object from the same LINCS platform, requiring no pathway database selection. H3 and H4 share this shRNA dependency (documented in DEVIATION_LOG.md entry 3) and are not independent tests. The phenotype-projected instability is applicable only to drugs with target

363 annotations—1,551 of 8,949 (17%). For the remaining 82% of drugs with unknown targets, the
364 validation framework provides other conditions (transport stability, localization) that do not require
365 target annotations.

366 As a pre-registered convergent-validity check, we recomputed phenotype-projected instability
367 using CRISPRi loss-of-function signatures [9] as the on-target direction. Projected instability cor-
368 relates with alignment in the same direction as the shRNA reference and more strongly (Section 5.4,
369 Table 3).

370 Under the pre-registered secondary criterion, raw instability correlates with CRISPRi alignment
371 at $\rho = -0.19$, so the sign pattern matches the shRNA arm rather than opposing it. The registered
372 criterion requires projected $|\rho| > 0.3$ and raw $|\rho| < 0.15$; the CRISPRi arm meets the first and not
373 the second, and so satisfies one of the two conditions. The shRNA arm satisfies both. CRISPRi
374 therefore supports H3 on the projected criterion while leaving the raw criterion unmet.

375 Because both compound and shRNA signatures were measured on the LINCS L1000 platform,
376 their association could reflect shared platform- or axis-specific structure. Raw instability does not
377 evaluate this threat because it does not use the target direction. The registered S2-T analysis
378 therefore tests whether the adjusted projected association exceeds that obtained after reassigning
379 complete target directions among targets.

380 5.4 Convergent validity across genetic references

381 We pre-registered a follow-up analysis of the two genetic references (frozen at commit 7f57136)
382 covering how the CRISPRi direction is built, how reliably each reference reproduces itself, whether
383 either distinguishes a drug’s annotated target from other targets, whether the association depends
384 on target composition, and whether broad cytotoxicity accounts for it. Table 3 reports the associ-
385 ations under four references.

386 The projected association is positive under every reference, and stronger under CRISPRi than
387 under shRNA. The raw association is negative under the CRISPRi references and not detectable
388 under shRNA. How the CRISPRi direction is built governs neither: rebuilding it from the genome-
389 wide screen rather than the essential screen moves $\rho(P, E)$ by 0.066 and $\rho(D, E)$ by -0.145 , and
390 both 95% intervals on the difference include zero.

391 **Reliability.** Split-half reliability of the reference direction, the median cosine between half-sample
392 consensus directions over 50 random splits, is 0.20 across 258 shRNA targets and 0.20 across 2,057
393 K562 CRISPRi targets. The RPE1 reference reaches 0.44 across 2,393 targets. Two of the three
394 references reproduce their own direction at a cosine near one fifth.

395 **Target identity.** Whether a reference separates a drug’s annotated target from other targets
396 is a different question from whether the association exists, and we registered it separately as the
397 mean percentile of a drug’s own-target alignment within its alignments to all other targets, where
398 0.5 is chance, against an equivalence region of $[0.4, 0.6]$. The shRNA reference returns 0.525 with
399 a 95% interval of $[0.501, 0.550]$, inside that region. The K562 CRISPRi reference returns 0.574
400 $[0.504, 0.644]$, which the registered rule reads as inconclusive. Only RPE1 returns a value the rule
401 reads as carrying target identity, 0.610 $[0.546, 0.671]$, at a permutation p of 7.0×10^{-4} . The ordering
402 follows reliability: the one reference that reproduces itself is the one that separates targets.

403 **Target composition.** The association is concentrated on essential-screen targets. Splitting the
404 genome-wide arm at membership in the K562 essential screen, with the other-target subset re-
405 stricted to targets carrying a transcriptional phenotype as registered, projected instability corre-

Table 3: Instability against signless target-axis alignment under four genetic references. P is phenotype-projected instability, D is raw direction instability, and $E = \cos^2(m, u)$ is signless alignment, which cannot separate alignment from anti-alignment. Brackets give cluster-bootstrap 95% intervals resampling targets. C0 and C1 are two constructions of the CRISPRi direction, from the deposited essential-screen signatures and from a rebuild on the genome-wide screen. Neither reference is a ground truth for target engagement.

Reference	Construction	Drugs	$\rho(P, E)$	$\rho(D, E)$
shRNA, LINCS	consensus	795	+0.317	-0.058
CRISPRi, K562	C0	131	+0.541 [0.341, 0.674]	-0.192 [-0.455, 0.108]
CRISPRi, K562	C1	131	+0.607 [0.350, 0.760]	-0.337 [-0.548, -0.086]
CRISPRi, RPE1	C1	131	+0.770 [0.650, 0.838]	-0.466 [-0.604, -0.302]

lates with alignment at $\rho = 0.658$ [0.442, 0.760] among 104 drugs over 33 essential-screen targets, and at $\rho = 0.098$ [-0.118, 0.280] among 182 drugs over 61 other targets. The difference, estimated by resampling targets within each subset independently, is 0.560 [0.278, 0.810] for the projected association and -0.373 [-0.680, -0.059] for the raw one, and both exclude zero. We do not claim that the association extends beyond essential-screen targets.

Cytotoxicity. On the 120 drugs with PRISM viability at a near-common exposure of 2–3 μM , raw instability correlates with alignment at $\rho = -0.363$ [-0.585, -0.109], and partialing viability out leaves -0.314 [-0.535, -0.044], an attenuation of -0.049 with a 95% interval of [-0.157, 0.031]. The adjustment is consistent with variation shared between viability and alignment, and does not establish that cytotoxicity produced the association.

5.5 Essential-gene correction (Experiment 7)

Universal-essential knockdowns have *higher* Grassmannian geodesic distances than non-essential knockdowns (mean 2.80 vs. 2.67, Cohen’s $d = +0.60$, Mann–Whitney $p < 10^{-33}$; $n = 1,567$ genes with both geodesic distances and DepMap labels, of which 855 universal-essential and 712 sub-threshold). Cohen’s $d = 0.60$ is a medium effect—the distributions overlap substantially, and individual genes span the full range in both groups. The *group means* differ in the opposite direction from the predicted confound, and that difference does not show that essentiality status determines geodesic distance.

Essential genes do not produce a shared response across cell types; they *diverge*—the same gene’s lethal disruption manifests through different transcriptomic programs in K562 and RPE1. This is consistent with cell-line-specific essentiality effects documented in CERES [16] and integrated Broad–Sanger analyses [17]: the functional consequences of losing a core-machinery gene depend on the cell’s lineage and metabolic context.

A label-permutation null (shuffling essential/non-essential labels across the 1,567 genes, 10,000 permutations) confirms that this divergence exceeds background K562/RPE1 divergence: the observed difference (0.123) falls outside the null 95% interval [-0.021, 0.022], $p_{\text{perm}} < 10^{-4}$. The null preserves the geodesic-distance distribution while breaking only the label assignment, so the $\sim 6\times$ exceedance of the null half-width rules out the library-overlap confound (85% target overlap between screens).

Spearman ρ between raw and corrected geodesic distances is 0.91 ($p < 10^{-100}$). The global ranking is preserved.

The correction’s value is at the item level. The top 100 rank-shifting genes are enriched for spliceosomal complex assembly (Fisher’s exact OR = 8.9, $p = 1.3 \times 10^{-7}$) and proteasomal catabolic processes (OR = 4.1, $p = 6.7 \times 10^{-5}$). Proteasome subunits (*PSMA3*, *PSMB6*, *PSMC4*) shift 750–900 rank positions; spliceosome components (*SF3B1*, *SNRPG*, *DDX23*) shift 660–820 positions. The group-mean comparison (essential vs. non-essential, Cohen’s $d = 0.60$) provides supporting context: the distributions overlap substantially, and essentiality status is a weak per-gene classifier. HP3 is best understood as a shift in group-level geometry localized to identifiable core-machinery complexes, not a gene-level discriminator. These complexes are the genes whose cross-cell-type geometry is most dominated by the shared essential-response axis; the correction provides a concrete action: flag them before interpreting their transport scores.

5.6 Morphological feature ablation (Experiment 8)

Removing the 252-feature morphological block left the compound ranking nearly unchanged (Spearman $\rho = 0.998$, $n = 24,992$).

Across 200 size-matched random feature ablations the median correlation was 0.99985 (2.5th–97.5th percentiles 0.99980–0.99988). The observed correlation, 0.99766, fell below all 200 null values, so the structured block perturbs the ranking more than deleting an equal number of arbitrary coordinates does. The displacement remains small in absolute terms, and the two readings are compatible: the ordering is preserved, and the block is not interchangeable with an arbitrary set of the same size.

Leave-one-source-out correlations ranged from 0.9975 to 0.9977 across all ten imaging sources, so no single site carries the result. Half the compounds moved less than 1.0 percentile point, 2.2% moved more than five, and the 90th and 99th percentiles of absolute percentile-rank shift were 3.2 and 5.9 points.

Two limits bound what this measures. Requiring observations from at least five sources retained 24,992 of 115,729 profiled compounds, so the result describes a well-covered subset and may not extend to sparsely profiled compounds. Only 11 of those 24,992 compounds map uniquely to PRISM Repurposing, so a compound-level cytotoxicity interpretation of the individual rank shifts cannot be evaluated on this cohort.

5.7 Summary of supplementary experiments

Full methods, tables, and discussion appear in Supplementary Experiments.

H1 (toxicity correction). Among 13 cytotoxic drugs, most are already high-instability ($D > 0.87$ for 9 of 13); toxicity correction shifts values by < 0.01 . The predicted failure mode—cytotoxicity masquerading as mechanism conservation—does not hold. Cytotoxicity produces genuinely context-dependent signatures in 978-gene space.

H2 (broad-mechanism drugs). Seven of 10 broad-mechanism drugs fall below the population median. Pan-HDAC inhibitors dominate ($D = 0.49$ – 0.55). The pre-registered second prediction—that kinase inhibitors would fall above the median—failed. The sorting axis is target breadth, not clinical specificity.

H4 (localization). Macro-averaged AUROC gap is +0.018 (below +0.05 threshold). Localization improves prediction for intracellular enzyme inhibitors (topoisomerase: +0.49, CDK: +0.26) but hurts for receptor-mediated drugs (acetylcholine: -0.20 , PPAR: -0.19). Condition 4 is domain-dependent.

HP2 (pathway function prediction). GO Jaccard overlap has near-zero correlation with geodesic distance ($|\rho| < 0.025$) for both raw and corrected forms. Geometric similarity of perturbation responses is distinct from functional pathway relatedness.

6 Discussion

6.1 Ranking preservation and item-level diagnosis

In practice, the framework changes which items proceed to follow-up. Failing a condition does not downgrade a drug’s ranking; it flags the drug for further investigation before a mechanistic claim is supported. Consider a concrete case:

Drug repurposing triage. Daunorubicin (topoisomerase II inhibitor, 19 cell lines) and MG-132 (proteasome inhibitor, 16 cell lines) have nearly identical raw direction instability ($D = 0.557$ vs. 0.565 , $\Delta = 0.008$). Raw direction instability ranks them equivalently—both in the mechanism-conserving regime. Phenotype-projected instability separates them: daunorubicin scores 11.1 with target-axis alignment 0.24 (its consistency aligns with *TOP2A* knockdown); MG-132 scores 3.5 with alignment < 0.001 (its consistency does not project onto *PSMB1* knockdown). One interpretation, consistent with proteasome-inhibitor pharmacology, is that MG-132 produces consistent transcriptional effects through broad proteostasis disruption rather than through the specific program of its annotated target subunit. A practitioner using raw direction instability alone would treat both drugs identically. After Condition 5 validation, daunorubicin is prioritized for mechanism-based follow-up; MG-132 requires investigation of what program drives its consistency. This illustrative pair was selected by matching raw direction instability ($|\Delta D| < 0.05$) and ranking by projected-instability divergence; individual drug-level scores carry uncertainty from cell-line sampling ($n = 16$ – 19).

6.2 What direction instability measures

Tower [4] finds that mechanism-of-action class explains 20% of residualized instability variance across 55,000 perturbations ($\eta^2 = 0.201$), separating core-machinery inhibitors from receptor-signaling drugs whose activity depends on target expression. A score varying with how widely a target is expressed, rather than with a removable nuisance axis, is one that feature and gene-set deletions should leave ordered. That is what the three domains show.

Under the DAG of Figure 2, cosine normalization blocks the linear magnitude path from perturbation strength to the metric. Direction instability operates in a specific regime: it compares signatures *across cell lines at a fixed dose*. In this regime, magnitude variation arises from cell-line-specific pharmacodynamics (how strongly each cell line responds), and cosine normalization removes it. The separability assumption—that signature direction is independent of response magnitude—holds trivially when the dose is fixed, because the “magnitude” that varies across cell lines is response amplitude, not concentration.

A stronger version of separability would require direction to be independent of dose concentration. A pre-registered empirical test (commit 5bb6080) assessed this stronger claim across 6,169 drug×cell-line pairs profiled at ≥ 2 doses (1,445 drugs, 52,920 dose-pair comparisons). The median within-pair cosine between dose-level signatures was 0.19, and 86.5% of pairs fell below 0.5—both pre-registered criteria failed (thresholds: median > 0.7 , $< 20\%$ below 0.5).

These low cosines are dominated by signature-level noise rather than genuine dose-dependent direction changes. Of dose-level groups in this analysis, 80% contain a single replicate; in 978 dimensions, two single-replicate z-score vectors have expected cosine near zero even when the

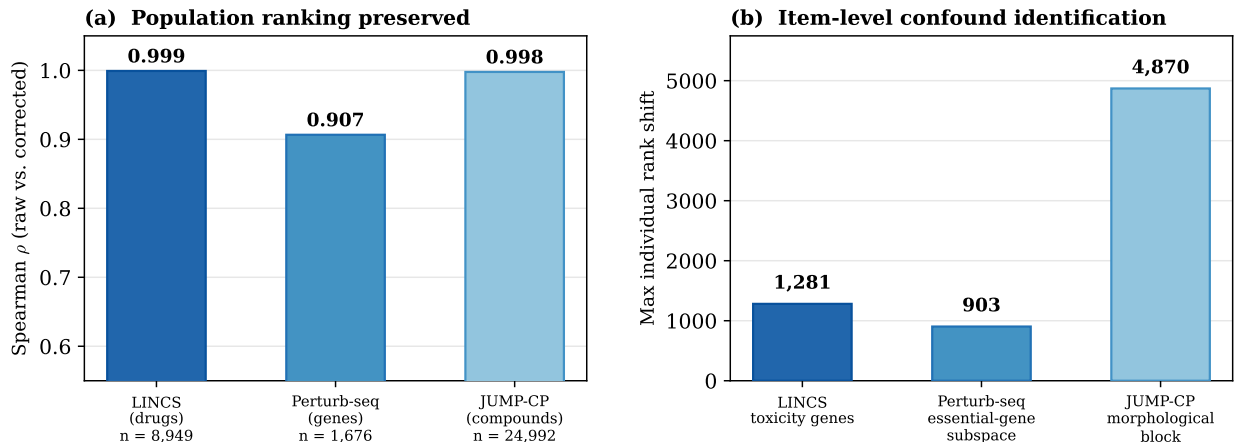


Figure 4: **Cross-domain empirical summary.** (a) Spearman ρ between raw and corrected scores in each domain. All exceed $\rho = 0.90$, so none of the corrections reorganizes the population ranking. Sample sizes shown below each bar. (b) Maximum individual rank shift per domain, showing that rank-preserving corrections still move specific items. In JUMP-CP the median absolute percentile-rank shift is 1.0 point and 2.2% of compounds move more than five, against a population $\rho = 0.998$.

underlying true direction is identical. A diagnostic control confirms this: near-identical doses (ratio < 1.01 , such as 9.99 vs. 10.00 μM —the same intended dose with measurement imprecision) yield median cosine = 0.11, establishing the noise floor. The overall median of 0.19 is barely above it. The pre-registered criteria cannot be met at this replicate depth, and the test lacks statistical power to assess separability.

The separability assumption therefore remains empirically unresolved: the test was underpowered for single-replicate data in 978 dimensions. A properly powered test would require ≥ 3 replicates per dose level and a within-dose replicate baseline—conditions that the LINCS Level 5 archive does not provide for most drug $\times$ cell $\times$ dose combinations.

Two additional boundary conditions affect the DAG’s blocking claim. Near-zero-norm signatures amplify noise under unit-normalization (guarded in code at $\|\mathbf{s}\| < 10^{-10}$). The Fréchet variance—the basis for the variance-regularized correction (Eq. 2)—is part of the DAG’s structure rather than a component of D itself. We have not proven identification with do-calculus; the DAG is a heuristic justification for cosine normalization as a confound-blocking adjustment, not a formal identification theorem.

6.3 Cross-domain consistency

The three analyses compare the sensitivity of geometric rankings to domain-specific adjustments, but they use different geometric estimands and context axes (Table 4). In LINCS, D measures directional consistency across cell lines. Perturb-seq compares per-cell-type response subspaces using Grassmannian geodesic distance. In JUMP-CP, D measures technical reproducibility across imaging sources. What the three share is that deleting the candidate nuisance dimensions preserves the global ranking. Item-level interpretation rests on an external criterion, which is available in the two transcriptomic domains and not in JUMP-CP.

Cytotoxic drugs occupy the high-instability regime, and universal-essential knockdowns produce divergent rather than shared response subspaces. Both results oppose the naive expectation that destructive perturbations generate a shared, apparently transportable response. The JUMP-

Table 4: Ablation sensitivity by domain. The table compares how much each feature or gene removal moves the population ranking; it does not assert that one construct is measured in all three. [†]JUMP-CP is an unregistered sensitivity analysis whose context axis is the imaging source, so its ρ concerns cross-site technical reproducibility rather than biological context-dependence. The HDAC calibration panel is omitted because the threshold was set from it. The LINCS HDAC gradient ρ measures fine-grained ordering within a single drug class (6 compounds in the selectivity gradient, Figure 1); the global 8,949-drug $\rho > 0.99$. H5 (transport stability) tests predictive validity rather than rank preservation; the 66-fold leave-one-out result (Table 2) provides the strongest domain-specific evidence for LINCS.

Domain	Correction	N	ρ
LINCS (global)	Toxicity genes	8,949	>0.99
Perturb-seq	Essential-gene subspace	1,676	0.91
JUMP-CP [†]	Morphological block	24,992	0.998

CP analysis instead measures the sensitivity of a cross-site technical-reproducibility ranking to morphological feature deletion.

6.4 Generalizability of the validation procedure

The validation framework was tested on one geometric perturbation score (direction instability) across three biological domains. The *procedure*—define interpretive threats, specify corrections that target each threat, pre-register predictions about whether corrections reorganize rankings, test empirically—applies to other geometric scores that measure perturbation-response consistency, such as connectivity scores [2], morphological similarity indices, and Grassmannian distance metrics. What would change are the domain-specific conditions: a connectivity score would require different confound corrections than direction instability, but the same structure of threat identification, correction, and pre-registered testing applies. The effect size for phenotype projection ($\rho = 0.32$, explaining $\sim 10\%$ of variance) is moderate. For comparison, connectivity-score benchmarks in LINCS typically report Spearman correlations of $\rho = 0.2$ – 0.4 between metric and biological outcome [18], placing the phenotype-projection result at the upper end of the range achievable with unsupervised geometric scores on transcriptomic data.

6.5 Scope and limitations

Direction instability measures angular consistency of perturbation signatures across contexts. It does not measure therapeutic efficacy, in-vivo translatability, or clinical specificity. The validation framework is a validation framework for perturbation-score interpretation, not a surrogate endpoint. The consequential validity facet (whether confidence upgrades change practitioner decisions) is illustrated via the daunorubicin/MG-132 case study but is not empirically tested; a user study or decision-simulation experiment would be needed to validate this facet.

The Perturb-seq analysis uses $K = 2$ cell types, providing no Fréchet variance, holonomy, or transport stability test. Both screens target DepMap-essential genes, so the HP3 “non-essential” group consists of sub-threshold essentials rather than genuinely non-essential genes. A genome-wide screen with genuine non-essential controls would be sharper, though the label-permutation null rules out the library-overlap confound.

The variance-regularized formula (Eq. 2) is a heuristic linear combination of direction instability and Fréchet variance, not a principled geometric operation on the signature manifold. The penalty weight $\lambda = 1.0$ was inherited from the companion paper [3] and was not optimized for this evaluation. A pre-registered sensitivity analysis across $\lambda \in [0.25, 5.0]$ found the registered win criterion met more often at larger λ . Because that criterion compares signed correlations between oppositely-oriented statistics (Deviation 6), the pattern reflects the orientation of the penalized score rather than its predictive value, and is not evidence of robustness.

LINCS L1000 profiles have known batch effects across plates and production sites. Batch-driven systematic direction differences would inflate direction instability for drugs profiled across batches, producing a confound not addressed by any validity condition tested here. LINCS measures 978 landmark genes, which restricts the toxicity gene set to that panel.

The Perturb-seq subspace dimension ($k = 5$) was fixed; geodesic distances depend on this choice, and a principled selection criterion (such as an explained-variance threshold) would strengthen the analysis.

The JUMP-CP block was defined by substring matching on CellProfiler names rather than by correlation with an independent viability assay, so it is a morphological feature family and not a validated cell-health signature; the analysis therefore does not establish or remove a cell-health confound. Its size-matched null samples individual columns uniformly, so the block’s larger effect may partly reflect correlation among jointly removed families. The cohort is also nearly disjoint from PRISM-annotated compound space: 11 of 24,992 compounds map uniquely, which precludes validating an item-level cytotoxicity reading of the rank shifts. The 78.4% attrition biases the retained set toward well-replicated compounds.

7 Conclusions

Direction instability predicts held-out cell-line agreement in all 66 leave-one-out folds (mean $\rho = -0.602$); a variance-regularized variant does not improve on it. Target-axis projection tracks signless alignment to the shRNA target axis ($\rho = 0.32$ vs. -0.06 for the raw score), and an independent CRISPRi reference tracks it in the same direction ($\rho = 0.54$). Essential-gene correction identifies proteasome/spliceosome genes as confound-dominated while preserving the population ranking ($\rho = 0.91$).

List of abbreviations

AUROC: area under the receiver operating characteristic curve; CRISPRi: CRISPR interference; DI: direction instability; GO: Gene Ontology; HDAC: histone deacetylase; JUMP-CP: Joint Undertaking for Morphological Profiling – Cell Painting; LINCS: Library of Integrated Network-Based Cellular Signatures; LOO: leave-one-out; MOA: mechanism of action; PCA: principal component analysis; shRNA: short hairpin RNA; TS: variance-regularized direction instability.

Declarations

Ethics approval and consent to participate

Not applicable. This study uses only publicly available datasets (LINCS L1000, Perturb-seq, JUMP-CP, DepMap) with no human subjects research.

Consent for publication

Not applicable.

Availability of data and materials

All data used in this study are publicly available: LINCS L1000 (GEO GSE92742), Perturb-seq [9], JUMP-CP [10], DepMap [27]. Code, pre-registration commits (SHA 8d74bd0 for initial freeze; SHA f3c88ed, tag prereg-extended-psb, for extended pre-registration; SHA 5bb6080 for dose-direction separability test), deviation log, and all experimental scripts are available at <https://github.com/elliotttower/direction-instability-drug-validity>. The repository was public before the pre-registration freeze. A permanent archive is deposited at Zenodo (DOI: <https://doi.org/10.5281/zenodo.21284088>).

Competing interests

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Authors' contributions

ET conceived the study, designed the validation framework, conducted all experiments, and wrote the manuscript.

Use of AI-assisted technologies

Claude (Anthropic) was used as a writing and analysis assistant during manuscript preparation. The author directed all analytical decisions and verified all numerical results against source data. The validation framework, pre-registered hypotheses, experimental results, and interpretations are the author's own. The author takes full responsibility for the content. No AI tool is listed as an author.

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Supplementary information

Additional file 1: Supplementary Experiments and Extended Results. Full methods, tables, and discussion for Experiments 1, 2, and 4 (H1 toxicity correction, H2 broad-mechanism drugs, H4 localization score) and HP2 (GO pathway prediction from corrected Perturb-seq distances). Includes the complete HDAC drug list used in the sensitivity analysis and Supplementary Table S1 (Messick validity-facet mapping).

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